

where the matrix $\mathbf{B}(\mathbf{z}) = (b_{ij}(\mathbf{z}))$ is related to $\mathbf{A}(\mathbf{z})$ by

$$\mathbf{B}(\mathbf{z}) = \Phi'^T(\mathbf{z})\mathbf{A}(\Phi(\mathbf{z}))\Phi'(\mathbf{z}).$$

We say that a 2-form ψ is *conserved under mapping* Φ if

$$\Phi^*\psi = \psi.$$

In coordinates, the conservation of the 2-form represented by matrix $\mathbf{A}$ under a mapping Φ means that

$$\Phi'^T \mathbf{A} \Phi' = \mathbf{A}.$$

In the particular case of the symplectic 2-form ψ_s , we have $\mathbf{A} = \mathbf{J}$, and the following condition for conservation under the mapping Φ

$$\Phi'^T \mathbf{J} \Phi' = \mathbf{J}. \quad (2.17)$$

A map that conserves the symplectic 2-form, or, in coordinates, satisfies (2.17), is termed a *symplectic map*.

Taking the determinant of both sides of (2.17), we have

$$\det(\Phi'^T \mathbf{J} \Phi') = \det(\mathbf{J}) \Rightarrow \det(\Phi'^T) \det(\mathbf{J}) \det(\Phi') = \det(\mathbf{J}),$$

hence

$$\det(\Phi')^2 = 1,$$

so $|\det(\Phi')| = 1$. This raises a curious issue since we could conceivably have $\det(\Phi') = -1$.

In the case of a flow map $\Phi = \mathcal{F}_t$, we know that for $t \rightarrow 0$, the map reduces to the identity map (the same has to happen for a consistent numerical method). This means that we would have to have

$$\lim_{t \rightarrow 0} \det(\mathcal{F}'_t) = 1.$$

If the system is Hamiltonian, the map is symplectic for all t , and the determinant will be a continuous function of t , so the cases of interest have $\det(\mathcal{F}'_t) = +1$. The flow map of a Hamiltonian system is volume preserving.